

LATOM Wellness Protocol: Insulin Resistance & Metabolic Optimization

A patient guide to reversing metabolic dysfunction using fiber, plant compounds, natural GLP-1 secretagogues, and lifestyle, with prescription support when needed

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WARNING: This protocol is educational. It does not establish a practitioner-patient relationship. For personalized recommendations, book a consultation at latomwellness.com.

Why insulin resistance matters even before you're diabetic

Insulin resistance is the metabolic condition behind type 2 diabetes, cardiovascular disease, fatty liver, polycystic ovary syndrome, Alzheimer's (sometimes called "type 3 diabetes"), and a meaningful portion of weight gain that resists diet and exercise. It begins decades before a diabetes diagnosis. Conventional care typically waits for an HbA1c to cross 5.7 percent (prediabetes) or 6.5 percent (diabetes) before acting. By that point you have been insulin resistant for 10 to 20 years.

You can find it earlier. You can reverse it earlier. And the same tools that reverse insulin resistance also drive weight loss, improve energy, sharpen cognition, and cut cardiovascular risk.

This protocol covers what to test, how to interpret it, and the evidence-based interventions including fiber, plant compounds, natural GLP-1 secretagogues, and lifestyle protocols. Prescription GLP-1 medications are covered briefly as an option when natural approaches plateau.

How to actually find insulin resistance (the labs that show what HbA1c misses)

THE FASTING INSULIN GAP

Standard primary care orders HbA1c and fasting glucose. Both are LATE markers. By the time HbA1c is elevated, insulin levels have been climbing for years to compensate.

Fasting insulin catches the problem 10 to 20 years earlier. Optimal is typically below 7 microIU/mL. Most

"healthy" people walking around have fasting insulin between 10 and 20. That is silent metabolic disease.

CORE LABS

Lab	What it tells you	Optimal range
Fasting insulin	Insulin compensation early signal	Below 7 microIU/mL
Fasting glucose	Standard glycemic marker	70 to 90 mg/dL
HOMA-IR (calculated from glucose and insulin)	Quantified insulin resistance	Below 1.5
HbA1c	3-month average glucose	Below 5.4 percent
Triglycerides	Hepatic insulin resistance marker	Below 80 mg/dL
Triglyceride to HDL ratio	Strongest practical IR marker	Below 1.5
Advanced lipid panel (NMR or ApoB + Lp(a))	Atherogenic particle count	ApoB below 80, Lp(a) below 50
High-sensitivity CRP (hsCRP)	Metabolic inflammation	Below 1.0 mg/L
Liver enzymes (ALT, AST, GGT) plus liver elasticity (FibroScan if available)	Fatty liver screen	ALT below 25 (women) or 30 (men)
Uric acid	Fructose load + IR marker	Below 5.5 mg/dL
Vitamin D 25-OH	Often deficient, affects insulin sensitivity	Above 50 ng/mL
TSH plus free T3 plus free T4 plus reverse T3 plus anti-TPO	Thyroid affects metabolism	Optimized, not just "normal"

ADJUNCT: CONTINUOUS GLUCOSE MONITOR (CGM)

Even in non-diabetic adults, wearing a CGM (Stelo, Lingo, Levels, Veri, or similar) for 2 to 4 weeks shows you exactly which foods spike your blood sugar and how badly. This is one of the most powerful behavior-change interventions available. Cost is roughly \$90 to \$200 for a 2-week wear.

WHAT THE NUMBERS MEAN TOGETHER

- Fasting insulin above 10 with HbA1c "normal" at 5.4 percent = early insulin resistance
- Triglyceride to HDL ratio above 2 = metabolic syndrome
- ApoB above 100 with insulin above 10 = high atherogenic risk
- ALT above 30 in a non-drinker = likely fatty liver
- These patterns together = full metabolic syndrome, even without a diabetes diagnosis

Tier 1: Fiber strategies that move the needle

Soluble viscous fibers slow gastric emptying, blunt glucose absorption, feed gut bacteria that produce

short-chain fatty acids, and trigger GLP-1 release naturally. These are the most evidence-backed dietary supplements for insulin resistance.

GLUCOMANNAN (KONJAC ROOT FIBER)

- **Dose:** 1 to 4 grams 15 to 30 minutes before meals with at least 8 ounces of water
- **Mechanism:** Forms a viscous gel in the stomach, delays gastric emptying, slows carbohydrate absorption, increases satiety, triggers ileal GLP-1 release
- **Evidence:** Multiple meta-analyses showing reductions in fasting glucose, HbA1c, total cholesterol, and body weight. FDA-approved health claim in some formulations.
(Source: Sood N, Baker WL, Coleman CI. Effect of glucomannan on plasma lipid and glucose concentrations, body weight, and blood pressure: systematic review and meta-analysis. Am J Clin Nutr. 2008;88(4):1167-1175.)
- **Cautions:** Must take with water to avoid esophageal obstruction risk. Start at 1 gram and titrate up. Can cause gas/bloating initially.

PSYLLIUM HUSK

- **Dose:** 5 to 10 grams (1 to 2 rounded tablespoons of whole husk, or equivalent capsules) before meals, 2 to 3 times daily
- **Mechanism:** Soluble fiber, gel-forming, slows glucose absorption, lowers LDL cholesterol via bile acid sequestration, supports stool quality
- **Evidence:** Well-established for both glycemic control and LDL lowering. Several meta-analyses confirm modest but consistent effects on HbA1c, fasting glucose, and total/LDL cholesterol.
(Source: Anderson JW, et al. Long-term cholesterol-lowering effects of psyllium as an adjunct to diet therapy in the treatment of hypercholesterolemia. Am J Clin Nutr. 2000;71(6):1433-1438.)
- **Cautions:** Take separately from medications (2-hour gap) to avoid binding. Start at 1 teaspoon and titrate. Drink plenty of water.

PARTIALLY HYDROLYZED GUAR GUM (PHGG / SUNFIBER)

- **Dose:** 5 to 10 grams daily, can be added to coffee or smoothie
- **Mechanism:** Soluble, fermentable, prebiotic. Gentler than psyllium for sensitive guts.
- **Evidence:** Used clinically in IBS, also has emerging evidence for glycemic control and gut health.

RESISTANT STARCH

- **Dose:** 10 to 30 grams daily from foods (cooked-and-cooled potatoes, green bananas, properly prepared legumes, oats), or 5 to 10 grams as raw potato starch
- **Mechanism:** Bypasses small intestinal digestion, ferments in the colon to short-chain fatty acids (especially butyrate), which improve gut barrier and insulin sensitivity
- **Evidence:** Emerging but increasingly robust. Improves insulin sensitivity in postprandial and chronic dosing studies.

BETA-GLUCAN (OAT OR BARLEY)

- **Dose:** 3 to 6 grams daily from oats or barley

- **Evidence:** FDA-recognized heart-health claim. Modest but consistent effects on cholesterol and post-meal glucose.
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Tier 1: Plant compounds with strong glucose evidence

BERBERINE

- **Dose:** 500 mg three times daily with meals, 12-week initial course
- **Mechanism:** AMPK activation, similar to metformin. Insulin sensitization, hepatic glucose output reduction, lipid lowering.
- **Evidence:** Several RCTs comparing berberine to metformin show comparable HbA1c reductions. Strong meta-analysis support.
(Source: Yin J, Xing H, Ye J. Efficacy of berberine in patients with type 2 diabetes mellitus. *Metabolism*. 2008;57(5):712-717.)
- **Cautions:** GI side effects (nausea, diarrhea) in 15 to 20 percent. Take with food. Avoid in pregnancy.

CHLOROGENIC ACID (FROM GREEN COFFEE BEAN OR YERBA MATE)

- **Dose:** 400 to 1000 mg standardized extract daily, or 1 to 3 cups of yerba mate (which contains both chlorogenic acid and theobromine)
- **Mechanism:** Inhibits glucose-6-phosphatase in the liver (reducing endogenous glucose production), slows intestinal glucose absorption, may stimulate GLP-1 release. Theobromine (also in yerba mate) modulates insulin sensitivity.
- **Evidence:** Studies on green coffee bean extract show modest reductions in fasting glucose and post-meal glucose response. Yerba mate trials show metabolic benefits, weight loss, and lipid improvements.
(Source: van Dijk AE, et al. Acute effects of decaffeinated coffee and the major coffee components chlorogenic acid and trigonelline on glucose tolerance. *Diabetes Care*. 2009;32(6):1023-1025.)
(Source: Bracesco N, et al. Recent advances on *Ilex paraguariensis* research: minireview. *J Ethnopharmacol*. 2011;136(3):378-384.)

CINNAMON (CEYLON, NOT CASSIA)

- **Dose:** 1 to 6 grams daily
- **Mechanism:** Insulin signaling modulation, slows gastric emptying
- **Evidence:** Modest effect on fasting glucose. Use Ceylon (true) cinnamon to avoid coumarin liver toxicity from Cassia.
- **Cautions:** Cassia cinnamon contains significant coumarin and can cause hepatotoxicity at sustained higher doses.

APPLE CIDER VINEGAR

- **Dose:** 1 to 2 tablespoons (15 to 30 mL) in a large glass of water before carbohydrate-containing meals
- **Mechanism:** Delays gastric emptying, blunts postprandial glucose spike
- **Evidence:** Multiple small RCTs show 20 to 40 percent reduction in post-meal glucose spikes.

- **Cautions:** Always dilute. Can erode tooth enamel if undiluted or sipped slowly.
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Tier 2: Natural GLP-1 secretagogues

GLP-1 (glucagon-like peptide-1) is the hormone that prescription drugs like semaglutide and tirzepatide mimic. Your body produces GLP-1 naturally from gut L-cells in response to food. Several natural compounds and dietary patterns increase endogenous GLP-1 release. These are not as potent as the prescription drugs but stack well, have no withdrawal rebound, and address the underlying mechanism rather than overriding it.

FOODS AND DIETARY PATTERNS THAT INCREASE ENDOGENOUS GLP-1

- **Protein-rich meals** (especially leucine-rich): whey isolate, eggs, fish, lean meat
- **High-fiber meals:** especially viscous fiber and resistant starch
- **Olive oil and monounsaturated fats**
- **Bitter compounds:** arugula, dandelion greens, radicchio, bitter melon, gentian root
- **Fermented foods:** short-chain fatty acid production drives GLP-1 release from L-cells

COMPOUNDS THAT MAY STIMULATE GLP-1 RELEASE

- **Yerba mate** (chlorogenic acid + saponins) - mate-saponins specifically shown to delay gastric emptying and increase satiety
- **Mulberry leaf extract (1-deoxynojirimycin / DNJ)** - alpha-glucosidase inhibitor, blunts post-meal glucose spike
- **Bitter melon (Momordica charantia)** - traditional use, modest modern evidence for fasting glucose
- **Hesperidin (citrus flavonoid)** - emerging evidence for endothelial function and glucose
- **Allulose** (a rare sugar replacement) - actively triggers GLP-1 release in human studies, useful as a sugar substitute

NOTABLE MENTION: THE YERBA MATE STACK

Three to four cups of yerba mate daily (especially before meals) delivers a moderate dose of chlorogenic acid, theobromine (caffeine cousin with calmer profile), saponins, and mate-saponins. Several trials show modest weight loss, lipid improvements, and post-meal glucose attenuation. Tastes pleasant once accustomed. Brew at 150 to 175 degrees F (not boiling) to preserve compounds and reduce bitterness.

Tier 2: Mineral and cofactor support

MAGNESIUM GLYCINATE

- **Dose:** 400 mg daily, typically before bed (also supports sleep, which independently affects insulin sensitivity)

- **Evidence:** Magnesium deficiency is associated with insulin resistance. Repletion modestly improves insulin sensitivity in deficient individuals.

CHROMIUM PICOLINATE

- **Dose:** 200 mcg daily
- **Evidence:** Modest effect on glucose, may reduce sweet cravings. Useful adjunct.

ALPHA-LIPOIC ACID (R-FORM PREFERRED)

- **Dose:** 600 mg daily
- **Evidence:** Improves insulin sensitivity, also neuroprotective. Often used in diabetic neuropathy.

INOSITOL (MYO-INOSITOL PLUS D-CHIRO-INOSITOL, 40:1 RATIO)

- **Dose:** 4 grams myo + 100 mg d-chiro daily
- **Evidence:** Strong evidence in PCOS-pattern insulin resistance (women). Improves ovulation, lowers fasting insulin and HbA1c. Emerging evidence in metabolic syndrome more broadly.

CREATINE MONOHYDRATE

- **Dose:** 5 grams daily
 - **Mechanism:** Often overlooked for insulin resistance. Supports lean mass (which is metabolically active), improves training capacity, recent evidence for direct glucose handling effects.
 - **Evidence:** RCTs in older adults and those with metabolic syndrome show favorable effects on glucose handling and body composition.
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Tier 3: Emerging and experimental

These have lower evidence levels but reasonable rationale. Use with informed-consent discussion about evidence strength.

- **Trans-resveratrol** (animal data strong, human data mixed)
 - **Curcumin** (anti-inflammatory, modest glucose data)
 - **NMN or NR (nicotinamide precursors)** (emerging human evidence)
 - **Sulforaphane (broccoli sprout extract)** (interesting preliminary data)
 - **Bitter melon extract**
 - **Spermidine** (food-based safer than supplement)
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Lifestyle protocol (non-optional)

The supplements above amplify the lifestyle interventions. They do not replace them.

TIME-RESTRICTED EATING

- 8 to 10 hour eating window most days (e.g., 11 AM to 7 PM, or 10 AM to 8 PM)
- 14 to 16 hour overnight fast
- Allows insulin to fall to baseline, improves hepatic insulin sensitivity
- Stricter (16 to 18 hour fasts) can accelerate results but are not necessary for most patients

EXERCISE (THE ORDER MATTERS)

- **Resistance training 3 to 4 days per week:** builds muscle, the body's largest glucose disposal site. Most underutilized intervention.
- **Zone 2 cardio 3 to 4 days per week, 45 to 60 minutes:** builds mitochondrial density, improves fat oxidation. Pace = able to hold conversation, slightly elevated heart rate (typically 60 to 70 percent max).
- **One 10-minute walk after each meal:** blunts postprandial glucose 20 to 30 percent. Free. Essentially mandatory.

SLEEP

- 7 to 9 hours, consistent timing
- One night of poor sleep can reduce next-day insulin sensitivity by 25 to 30 percent
- This single intervention often outperforms supplements

STRESS MANAGEMENT

- Chronic stress drives cortisol, which drives insulin resistance
- Vagal tone work, breathwork, regular nature exposure, social connection
- Address chronic stress with the same seriousness as diet

DIET COMPOSITION

- Whole-food carbohydrates: vegetables, legumes, low-glycemic fruits, modest portions of whole grains
- Adequate protein: 0.7 to 1 gram per pound of lean body mass daily
- Healthy fats: olive oil, fatty fish, nuts, seeds
- Eliminate: ultra-processed foods, sugar-sweetened beverages, industrial seed oils in excess
- Test individual carbohydrate tolerance with CGM data

When to consider prescription GLP-1 (semaglutide / tirzepatide)

Prescription GLP-1 agonists work. They are not magic and they are not for everyone. Consider them when:

- HbA1c is in diabetic range (above 6.5) and lifestyle interventions have failed
- BMI is above 30 with comorbidities, or above 35, and lifestyle has failed
- Patient has obesity-related complications (sleep apnea, fatty liver, joint disease)
- You have ruled out underlying drivers (thyroid, sleep, food sensitivities, medication-induced weight gain)

IMPORTANT CONTEXT

- GLP-1 medications work while you take them. Weight regain is common when stopped without a transition plan.
- The natural approaches in this protocol build the underlying machinery that GLP-1 medications work through. They are not redundant. They are foundational.
- Compounded versus brand-name GLP-1s are a separate conversation with FDA, insurance, and supply considerations.

HOW LATOM APPROACHES THIS

We focus on lab-driven, lifestyle-first, supplement-supported protocols. Prescription GLP-1 is available for patients who need it, after appropriate evaluation. We are not a "Hims for GLP-1" practice. If you are looking for a fast online GLP-1 prescription with no workup, we are not the right fit.

When to escalate beyond this protocol

Refer to endocrinology or cardiology if:

- HbA1c above 9 percent (poorly controlled diabetes)
 - Symptoms of severe hyperglycemia (polyuria, polydipsia, weight loss, vision changes)
 - New cardiovascular symptoms (chest pain, syncope, severe shortness of breath)
 - Hypoglycemic episodes
 - Suspected pituitary, adrenal, or thyroid disease beyond garden-variety hypothyroid
 - Pediatric or pregnancy cases
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How to work with LATOM on this

STEP 1: BOOK A CONSULTATION

30-minute video visit. We review your symptoms, history, prior labs, current medications, and goals.

STEP 2: TARGETED LABS

Typically fasting insulin, HbA1c, HOMA-IR, advanced lipid (ApoB + Lp(a)), hsCRP, liver enzymes, uric acid, full thyroid, vitamin D. Plus CGM if desired.

STEP 3: PROTOCOL BUILD

Based on results, we build a stepwise plan:

- Fiber strategy matched to your tolerance
- Plant compound supplements through our Fullscript dispensary
- Lifestyle plan: time-restricted eating, exercise programming, sleep hygiene

- Optional CGM coaching
- Prescription bridge (GLP-1, metformin, or other) where clinically appropriate after lifestyle trial

STEP 4: FOLLOW-UP

12-week check-in with repeat labs. Adjust protocol based on response.

Cost expectations

Item	Cost range
LATOM consultation (initial)	\$149
Lab panel (insulin, HbA1c, advanced lipid, hsCRP, liver, thyroid, D)	\$150 to \$300
CGM (2-week wear)	\$90 to \$200
Supplement starter stack (30 days)	\$80 to \$180
Total first 30 days (without Rx)	\$469 to \$829

Prescription GLP-1 cost varies dramatically: \$300 to \$1,200 monthly for brand-name, \$150 to \$500 monthly for compounded (where legally available). Not included in the above.

What this protocol is NOT

- Not a substitute for emergency or specialist care
 - Not appropriate during active pregnancy without OB approval
 - Not a guarantee that all metabolic dysfunction can be fixed without prescription medication
 - Not a fast-pass GLP-1 prescription service. We require workup, lifestyle commitment, and informed consent.
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Sources

1. Sood N, Baker WL, Coleman CI. Effect of glucomannan on plasma lipid and glucose concentrations, body weight, and blood pressure: systematic review and meta-analysis. Am J Clin Nutr. 2008;88(4):1167-1175.
2. Anderson JW, Allgood LD, Lawrence A, et al. Cholesterol-lowering effects of psyllium intake adjunctive to diet therapy. Am J Clin Nutr. 2000;71(6):1433-1438.
3. Yin J, Xing H, Ye J. Efficacy of berberine in patients with type 2 diabetes mellitus. Metabolism.

2008;57(5):712-717.

4. van Dijk AE, Olthof MR, Meeuse JC, et al. Acute effects of decaffeinated coffee and the major coffee components chlorogenic acid and trigonelline on glucose tolerance. *Diabetes Care*. 2009;32(6):1023-1025.

5. Bracesco N, Sanchez AG, Contreras V, Menini T, Gugliucci A. Recent advances on *Ilex paraguariensis* research: minireview. *J Ethnopharmacol*. 2011;136(3):378-384.

6. Cani PD, Knauf C, Iglesias MA, Drucker DJ, Delzenne NM, Burcelin R. Improvement of glucose tolerance and hepatic insulin sensitivity by oligofructose requires a functional glucagon-like peptide 1 receptor. *Diabetes*. 2006;55(5):1484-1490.

7. Unfer V, Carlomagno G, Dante G, Facchinetti F. Effects of myo-inositol in women with PCOS: a systematic review of randomized controlled trials. *Gynecol Endocrinol*. 2012;28(7):509-515.

8. Hallsworth K, Adams LA. Lifestyle modification in NAFLD/NASH: Facts and figures. *JHEP Rep*. 2019;1(6):468-479.

This educational guide is provided by LATOM Wellness, PLLC. For personalized care, book a consultation at latomwellness.com.

Information in this protocol reflects clinical evidence as of 2026-05-24 and is subject to change as new research emerges. Re-verify dosing, contraindications, and drug interactions with your prescribing physician.